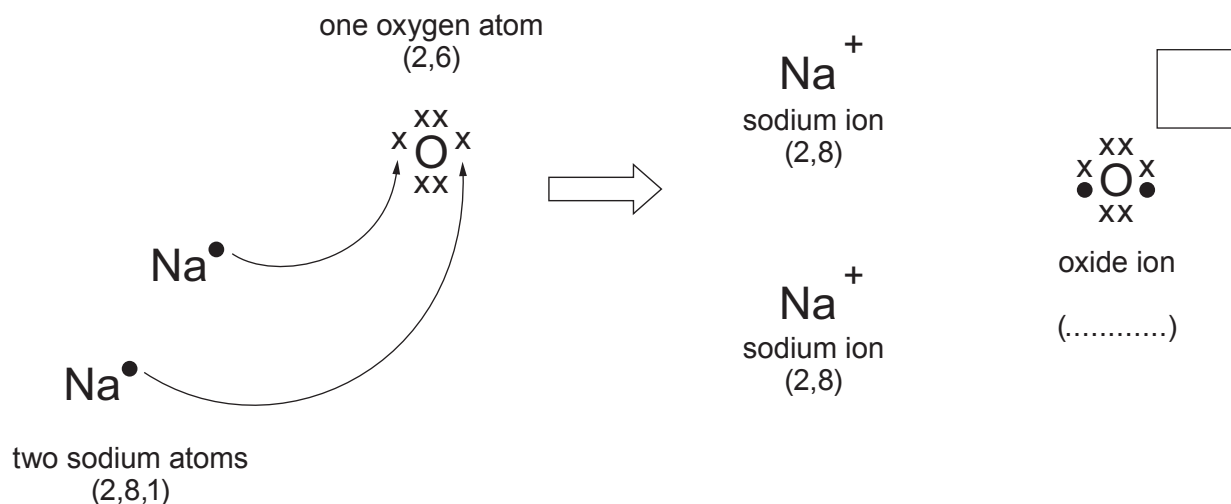


4. (a) The diagram shows the electronic changes that occur when sodium reacts with oxygen to form sodium oxide. The ● and x symbols are outer shell electrons.



- (i) **Complete the diagram** by putting in the electronic structure **and** the charge of the oxide ion. [2]
- (ii) Complete and balance the **symbol** equation for the reaction between sodium and oxygen. [2]



- (b) The table shows the electronic structure of the elements present in water and hydrogen chloride.

| Element  | Electronic structure |
|----------|----------------------|
| hydrogen | 1                    |
| oxygen   | 2,6                  |
| chlorine | 2,8,7                |

The diagram shows the bonding in a water molecule.



Give the **letter** of the diagram which shows the bonding in a hydrogen chloride molecule. [1]



A



B



C

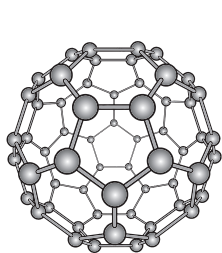


D

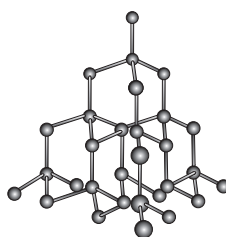
Letter .....



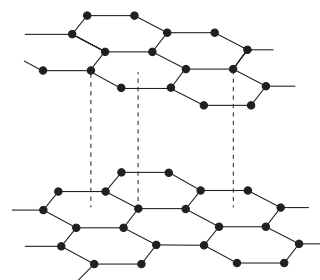
(c) Fullerene, diamond and graphite are different forms of carbon.



fullerene



diamond



graphite

(i) Underline the structure of these three forms of carbon.

[1]

**simple molecular**

**giant covalent**

**giant ionic**

(ii) The box shows some properties of different forms of carbon.

|                    |                           |                                     |
|--------------------|---------------------------|-------------------------------------|
| <b>transparent</b> | <b>insoluble in water</b> | <b>conducts electricity</b>         |
| <b>hollow</b>      | <b>soft</b>               | <b>does not conduct electricity</b> |
|                    |                           | <b>hard</b>                         |

Complete the table by choosing the property which makes the different forms of carbon suitable for the uses shown. [3]

| Form of carbon | Use           | Property |
|----------------|---------------|----------|
| diamond        | drill bits    |          |
| graphite       | pencils       |          |
| fullerene      | drug delivery |          |

